

PhysLive: 100 Simulation Integration Evidence

Automated diversity probe for the current schema-to-scene pipeline

Result: **100/100 cases passed**. Every selected schema produced a non-empty compiled scene graph and passed the frontend scene validation boundary. The probe covered nine curriculum topics and used a round-robin selection to avoid testing only one domain.

Metric	Result
Requested cases	100
Executed cases	100
Passed	100
Failed	0
Topics	9
Evidence captures	10 sheets x 10 cases

What this proves

This is an integration and renderability probe. It proves that the current frontend can compile the selected backend schema contracts into renderable scenes, including fallback scenes, graph series, semantic props, vector scenes, and scalar-field wave scenes. It does not replace numerical solver accuracy tests or an end-to-end test with every possible natural-language prompt.

Test correction discovered

The first run found one real contract issue in `water_surface_interference`: its authored `waveField` did not provide `probeX`. The frontend now supplies the safe physical-origin default `probeX = 0` while preserving an authored value when present. The rerun passed all 100 cases.

Coverage by topic

Topic	Cases	Pass	Example schema
CIRCUITS	12	12	thermistor_response
DYNAMICS	12	12	linear_drag_motion
ELECTROMAGNETISM	9	9	uniform_electric_field
KINEMATICS	12	12	kinematics
MODERN_PHYSICS	12	12	eclipse_geometry
OPTICS	12	12	thin_lens_imaging
PRACTICAL_AND_DATA	7	7	energy_environment
THERMAL	12	12	ideal_gas_isothermal
WAVES	12	12	radio_signal_chain

Evidence capture sheets

Each sheet below contains ten independently compiled simulation cases. The small canvas panels summarize the actual scene graph output: node count, prop count, graph trace, and pass status. The raw machine-readable result is stored at `tmp/simulation-probe/results.json`.

CASE 001

PASS

thermistor_response
CIRCUITS



nodes 8 props 2

CASE 002

PASS

linear_drag_motion
DYNAMICS



nodes 11 props 1

CASE 003

PASS

uniform_electric_field
ELECTROMAGNETISM



nodes 8 props 2

CASE 004

PASS

kinematics
KINEMATICS



nodes 13 props 0

CASE 005

PASS

eclipse_geometry
MODERN_PHYSICS



nodes 8 props 1

CASE 006

PASS

thin_lens_imaging
OPTICS

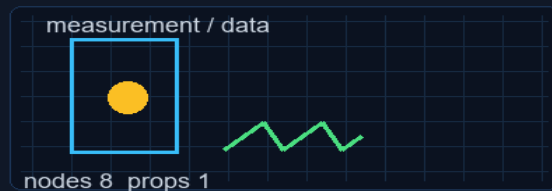


nodes 7 props 1

CASE 007

PASS

energy_environment
PRACTICAL_AND_DATA

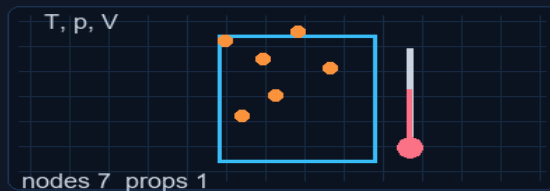


nodes 8 props 1

CASE 008

PASS

ideal_gas_isothermal
THERMAL



nodes 7 props 1

CASE 009

PASS

radio_signal_chain
WAVES



nodes 8 props 1

CASE 010

PASS

source_internal_resistance
CIRCUITS



nodes 9 props 2

CASE 011

PASS

moment_equilibrium
DYNAMICS



CASE 012

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 013

PASS

kinematics_projectile
KINEMATICS



CASE 014

PASS

energy_band_transition
MODERN_PHYSICS



CASE 015

PASS

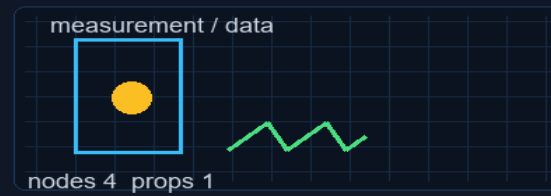
snell_refraction
OPTICS



CASE 016

PASS

measurement_uncertainty
PRACTICAL_AND_DATA



CASE 017

PASS

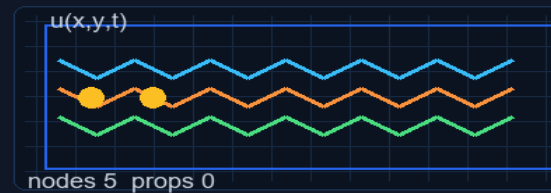
calorimetry_mixing
THERMAL



CASE 018

PASS

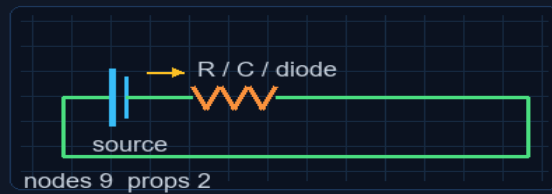
string_wave
WAVES



CASE 019

PASS

sensor_op_amp
CIRCUITS



CASE 020

PASS

dynamics_forces
DYNAMICS



CASE 021

PASS

point_charge_field
ELECTROMAGNETISM



CASE 022

PASS

kinematics_position_time_...
KINEMATICS



CASE 023

PASS

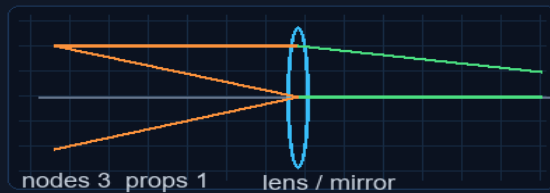
nuclear_reaction_energy
MODERN_PHYSICS



CASE 024

PASS

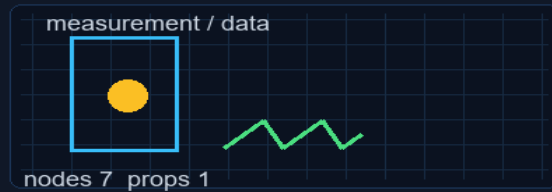
light_interference
OPTICS



CASE 025

PASS

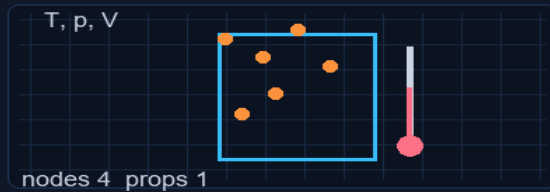
experimental_data_graph
PRACTICAL_AND_DATA



CASE 026

PASS

first_law_thermodynamics
THERMAL



CASE 027

PASS

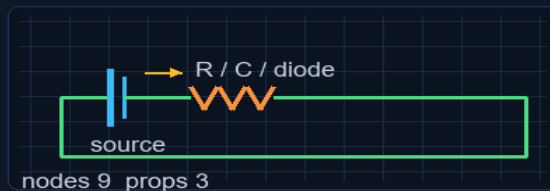
wave_pulse
WAVES



CASE 028

PASS

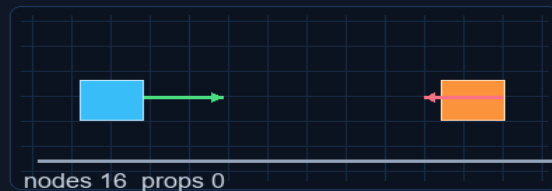
circuits_rc_charging
CIRCUITS



CASE 029

PASS

dynamics_collision
DYNAMICS



CASE 030

PASS

magnetic_force
ELECTROMAGNETISM



CASE 031

PASS

kinematics_velocity_time_...
KINEMATICS



CASE 032

PASS

radioactive_decay
MODERN_PHYSICS



CASE 033

PASS

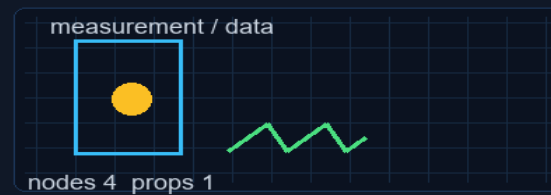
diffraction_polarization
OPTICS



CASE 034

PASS

measurement_uncertainty
PRACTICAL_AND_DATA



CASE 035

PASS

temperature_scales
THERMAL



CASE 036

PASS

standing_wave
WAVES



CASE 037

PASS

circuits_rc_discharging
CIRCUITS



CASE 038

PASS

oscillations_spring
DYNAMICS



CASE 039

PASS

uniform_electric_field
ELECTROMAGNETISM



CASE 040

PASS

kinematics_acceleration_t...
KINEMATICS

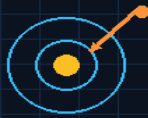


CASE 041

PASS

photoelectric_effect
MODERN_PHYSICS

atom / photon / bands

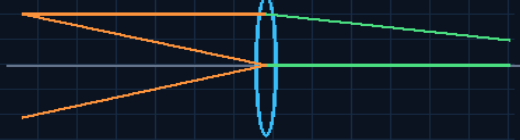


nodes 4 props 1

CASE 042

PASS

simple_magnifier
OPTICS



nodes 1 props 0 lens / mirror

CASE 043

PASS

energy_environment
PRACTICAL_AND_DATA

measurement / data



nodes 8 props 1

CASE 044

PASS

thermal_expansion
THERMAL

T, p, V



nodes 3 props 1

CASE 045

PASS

wave_reflection
WAVES

$u(x,y,t)$

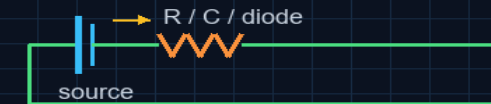


nodes 5 props 0

CASE 046

PASS

ac_rlc_circuit
CIRCUITS



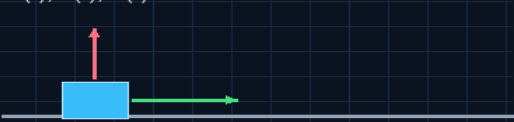
nodes 5 props 2

CASE 047

PASS

work_energy_power
DYNAMICS

$x(t)$, $v(t)$, $a(t)$



nodes 7 props 1

CASE 048

PASS

magnetic_force
ELECTROMAGNETISM



nodes 5 props 2

CASE 049

PASS

kinematics
KINEMATICS

$x(t)$, $v(t)$, $a(t)$



nodes 13 props 0

CASE 050

PASS

nuclear_energy
MODERN_PHYSICS

atom / photon / bands



nodes 5 props 2

CASE 051

PASS

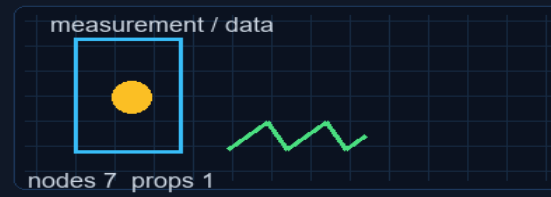
compound_microscope
OPTICS



CASE 052

PASS

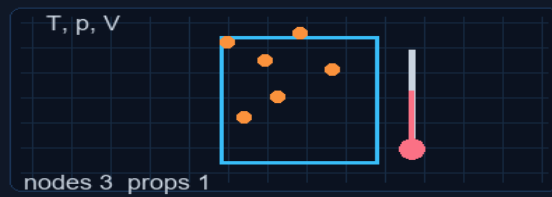
experimental_data_graph
PRACTICAL_AND_DATA



CASE 053

PASS

adiabatic_gas
THERMAL



CASE 054

PASS

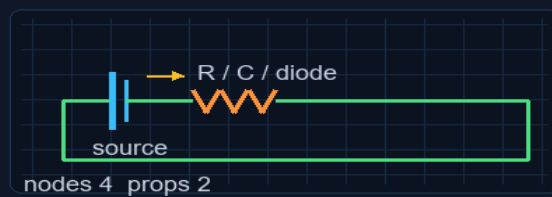
wave_superposition
WAVES



CASE 055

PASS

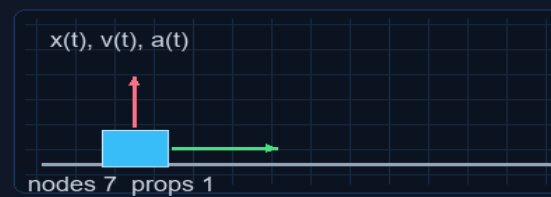
ohms_law
CIRCUITS



CASE 056

PASS

circular_motion
DYNAMICS



CASE 057

PASS

point_charge_field
ELECTROMAGNETISM



CASE 058

PASS

kinematics_projectile
KINEMATICS



CASE 059

PASS

atomic_spectra
MODERN_PHYSICS



CASE 060

PASS

astronomical_telescope
OPTICS



CASE 061

PASS

measurement_uncertainty
PRACTICAL_AND_DATA

measurement / data



CASE 062

PASS

ideal_gas_isobaric
THERMAL

T, p, V

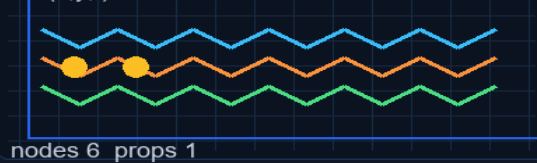


CASE 063

PASS

sound_wave
WAVES

$u(x,y,t)$



CASE 064

PASS

resistors_series
CIRCUITS



CASE 065

PASS

hooke_law
DYNAMICS

$x(t), v(t), a(t)$



CASE 066

PASS

electromagnetic_induction
ELECTROMAGNETISM



CASE 067

PASS

kinematics_position_time_...
KINEMATICS

$x(t), v(t), a(t)$

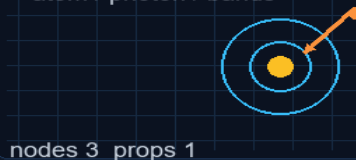


CASE 068

PASS

radiation_safety
MODERN_PHYSICS

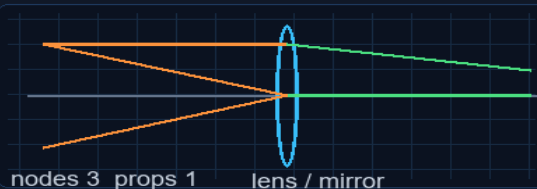
atom / photon / bands



CASE 069

PASS

diffraction_polarization
OPTICS



CASE 070

PASS

ideal_gas_isochoric
THERMAL

T, p, V



CASE 071

PASS

water_surface_interference
WAVES



CASE 072

PASS

resistors_parallel
CIRCUITS



CASE 073

PASS

gravity_orbit
DYNAMICS



CASE 074

PASS

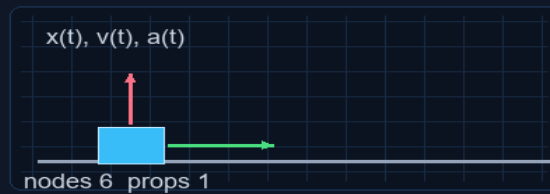
uniform_electric_field
ELECTROMAGNETISM



CASE 075

PASS

kinematics_velocity_time_...
KINEMATICS



CASE 076

PASS

xray_imaging
MODERN_PHYSICS



CASE 077

PASS

light_interference
OPTICS



CASE 078

PASS

phase_change
THERMAL



CASE 079

PASS

radio_communication
WAVES



CASE 080

PASS

capacitor_basic
CIRCUITS



CASE 081

PASS

hydrostatics
DYNAMICS



CASE 082

PASS

kinematics_acceleration_t...
KINEMATICS



CASE 083

PASS

ct_reconstruction
MODERN_PHYSICS



CASE 084

PASS

simple_magnifier
OPTICS



CASE 085

PASS

adiabatic_gas
THERMAL



CASE 086

PASS

ultrasound_imaging
WAVES



CASE 087

PASS

ac_waveform
CIRCUITS



CASE 088

PASS

damped_forced_oscillation
DYNAMICS



CASE 089

PASS

kinematics
KINEMATICS



CASE 090

PASS

mri_relaxation
MODERN_PHYSICS



CASE 091

PASS

compound_microscope
OPTICS



CASE 092

PASS

thermal_expansion
THERMAL



CASE 093

PASS

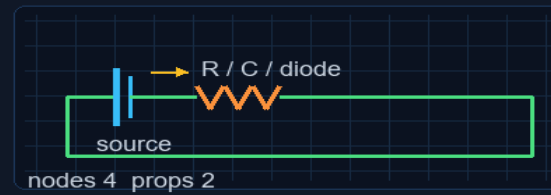
sound_wave
WAVES



CASE 094

PASS

ideal_transformer
CIRCUITS



CASE 095

PASS

gravity_orbit
DYNAMICS



CASE 096

PASS

kinematics_projectile
KINEMATICS



CASE 097

PASS

de_broglie_diffraction
MODERN_PHYSICS



CASE 098

PASS

astronomical_telescope
OPTICS



CASE 099

PASS

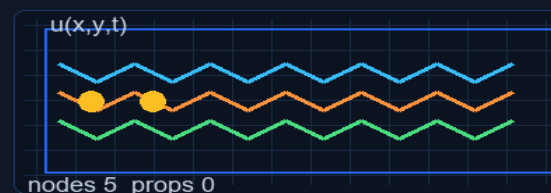
ideal_gas_isochoric
THERMAL



CASE 100

PASS

standing_wave
WAVES



Appendix: case matrix

#	Schema	Topic	Nodes	Props	Status
1	thermistor_response	CIRCUITS	8	2	PASS
2	linear_drag_motion	DYNAMICS	11	1	PASS
3	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
4	kinematics	KINEMATICS	13	0	PASS
5	eclipse_geometry	MODERN_PHYSICS	8	1	PASS
6	thin_lens_imaging	OPTICS	7	1	PASS
7	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
8	ideal_gas_isothermal	THERMAL	7	1	PASS
9	radio_signal_chain	WAVES	8	1	PASS
10	source_internal_resistance	CIRCUITS	9	2	PASS
11	moment_equilibrium	DYNAMICS	8	1	PASS
12	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
13	kinematics_projectile	KINEMATICS	14	2	PASS
14	energy_band_transition	MODERN_PHYSICS	9	2	PASS
15	snell_refraction	OPTICS	4	1	PASS
16	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
17	calorimetry_mixing	THERMAL	4	1	PASS
18	string_wave	WAVES	5	0	PASS
19	sensor_op_amp	CIRCUITS	9	2	PASS
20	dynamics_forces	DYNAMICS	13	0	PASS
21	point_charge_field	ELECTROMAGNETISM	5	2	PASS
22	kinematics_position_time_...	KINEMATICS	6	1	PASS
23	nuclear_reaction_energy	MODERN_PHYSICS	8	2	PASS
24	light_interference	OPTICS	3	1	PASS
25	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS
26	first_law_thermodynamics	THERMAL	4	1	PASS
27	wave_pulse	WAVES	5	0	PASS
28	circuits_rc_charging	CIRCUITS	9	3	PASS
29	dynamics_collision	DYNAMICS	16	0	PASS
30	magnetic_force	ELECTROMAGNETISM	5	2	PASS
31	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
32	radioactive_decay	MODERN_PHYSICS	6	1	PASS
33	diffraction_polarization	OPTICS	3	1	PASS
34	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
35	temperature_scales	THERMAL	3	1	PASS
36	standing_wave	WAVES	5	0	PASS
37	circuits_rc_discharging	CIRCUITS	9	3	PASS
38	oscillations_spring	DYNAMICS	12	1	PASS

#	Schema	Topic	Nodes	Props	Status
39	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
40	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
41	photoelectric_effect	MODERN_PHYSICS	4	1	PASS
42	simple_magnifier	OPTICS	1	0	PASS
43	energy_environment	PRACTICAL_AND_DATA	8	1	PASS
44	thermal_expansion	THERMAL	3	1	PASS
45	wave_reflection	WAVES	5	0	PASS
46	ac_rlc_circuit	CIRCUITS	5	2	PASS
47	work_energy_power	DYNAMICS	7	1	PASS
48	magnetic_force	ELECTROMAGNETISM	5	2	PASS
49	kinematics	KINEMATICS	13	0	PASS
50	nuclear_energy	MODERN_PHYSICS	5	2	PASS
51	compound_microscope	OPTICS	1	0	PASS
52	experimental_data_graph	PRACTICAL_AND_DATA	7	1	PASS
53	adiabatic_gas	THERMAL	3	1	PASS
54	wave_superposition	WAVES	5	0	PASS
55	ohms_law	CIRCUITS	4	2	PASS
56	circular_motion	DYNAMICS	7	1	PASS
57	point_charge_field	ELECTROMAGNETISM	5	2	PASS
58	kinematics_projectile	KINEMATICS	14	2	PASS
59	atomic_spectra	MODERN_PHYSICS	4	2	PASS
60	astronomical_telescope	OPTICS	1	0	PASS
61	measurement_uncertainty	PRACTICAL_AND_DATA	4	1	PASS
62	ideal_gas_isobaric	THERMAL	7	1	PASS
63	sound_wave	WAVES	6	1	PASS
64	resistors_series	CIRCUITS	4	2	PASS
65	hooke_law	DYNAMICS	7	1	PASS
66	electromagnetic_induction	ELECTROMAGNETISM	5	2	PASS
67	kinematics_position_time_...	KINEMATICS	6	1	PASS
68	radiation_safety	MODERN_PHYSICS	3	1	PASS
69	diffraction_polarization	OPTICS	3	1	PASS
70	ideal_gas_isochoric	THERMAL	7	1	PASS
71	water_surface_interference	WAVES	3	1	PASS
72	resistors_parallel	CIRCUITS	4	2	PASS
73	gravity_orbit	DYNAMICS	7	1	PASS
74	uniform_electric_field	ELECTROMAGNETISM	8	2	PASS
75	kinematics_velocity_time_...	KINEMATICS	6	1	PASS
76	xray_imaging	MODERN_PHYSICS	7	1	PASS
77	light_interference	OPTICS	3	1	PASS

#	Schema	Topic	Nodes	Props	Status
78	phase_change	THERMAL	8	1	PASS
79	radio_communication	WAVES	7	1	PASS
80	capacitor_basic	CIRCUITS	5	3	PASS
81	hydrostatics	DYNAMICS	7	1	PASS
82	kinematics_acceleration_t...	KINEMATICS	6	1	PASS
83	ct_reconstruction	MODERN_PHYSICS	7	1	PASS
84	simple_magnifier	OPTICS	1	0	PASS
85	adiabatic_gas	THERMAL	3	1	PASS
86	ultrasound_imaging	WAVES	7	1	PASS
87	ac_waveform	CIRCUITS	4	2	PASS
88	damped_forced_oscillation	DYNAMICS	10	1	PASS
89	kinematics	KINEMATICS	13	0	PASS
90	mri_relaxation	MODERN_PHYSICS	7	1	PASS
91	compound_microscope	OPTICS	1	0	PASS
92	thermal_expansion	THERMAL	3	1	PASS
93	sound_wave	WAVES	6	1	PASS
94	ideal_transformer	CIRCUITS	4	2	PASS
95	gravity_orbit	DYNAMICS	7	1	PASS
96	kinematics_projectile	KINEMATICS	14	2	PASS
97	de_broglie_diffraction	MODERN_PHYSICS	8	2	PASS
98	astronomical_telescope	OPTICS	1	0	PASS
99	ideal_gas_isochoric	THERMAL	7	1	PASS
100	standing_wave	WAVES	5	0	PASS